

TemporalGraph-ToM: Evaluating Theory-of-Mind Reasoning with Temporal Belief Graphs

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Extended Abstract

Higher-order theory of mind (ToM), the ability to attribute mental states to others, including their beliefs about others’ beliefs, remains a consistent challenge in large language model reasoning. Existing benchmarks evaluate higher-order belief recursion, such as Hi-ToM [1], interactional information asymmetry in FANToM [2], broad social-cognitive ToM abilities in ToMBench [3], first-person social and counterfactual reasoning in EgoSocialArena [4], and comprehensive cognition-inspired ToM paradigms in CogToM [5].

We introduce TemporalGraph-ToM (TGToM), a benchmark of 1k multi-agent stories that probes higher-order theory of mind with temporal belief graphs. TGToM uses controlled simulations up to seventh-order belief attribution, with information asymmetry from public and private claims, observation asymmetry from interleaved exits, agent reasoning involving deception and cooperation, and higher-order counterfactual scenarios. Evaluation questions span world state, higher-order beliefs, causal beliefs, common knowledge, and counterfactual beliefs.

Probing the limits of higher-order reasoning is relevant to LLM safety, agent safety, human-AI interaction, and reasoning interpretability. In evaluations of frontier models, we find a reality-bias (omniscient reasoning) failure mode: models default to the actual world state even when an agent’s belief diverges from it. TGToM evaluates whether models can correctly attribute belief updates under a belief propagation rule and exposes misattribution when models confuse the true world state with an agent’s propagated belief.

Reasoning scaffold. We introduce Temporal Belief Graph (TBG) and compare it with CoT and System 2 prompting. TBG is formalized as a temporal multilayer graph where nodes represent each agent’s state updated after an event. Beliefs propagate through four directed link types in two layers:

communication links (trusted claim, untrusted claim) and intent links (cooperative, deceptive). Beliefs additionally update directly when an agent observes an event. The node states and link types are generalizable to multiple beliefs, other mental states, event types, and agent interactions.

Temporal Belief Graph

Temporal belief graph is formalized as a temporal multilayer graph where nodes represent each agent's state (belief, intent, observation) updated after an event. Node state for agent i at time t :

$$s_i^{(t)} = \left(b_i^{(t)}, \iota_i^{(t)}, o_i^{(t)} \right)$$

where:

- $b_i^{(t)}$: belief
- $\iota_i^{(t)}$: intent
- $o_i^{(t)}$: observation

Beliefs propagate through four directed link types in two layers: communication links (trusted claim, untrusted claim) and intent links (cooperative, deceptive). Beliefs additionally update directly when an agent observes an event.

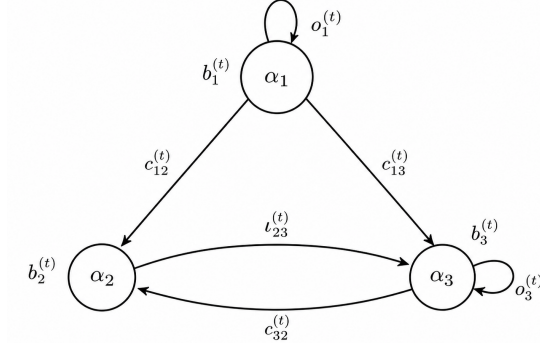


Figure 1: Temporal belief graph for three nodes. Node state $b_i^{(t)}$ denotes agent i 's belief at time t ; self-updates $o_i^{(t)}$ denote belief updates through observations. Directed edge labels $c_{ij}^{(t)}$ and $\iota_{ij}^{(t)}$ denote belief propagation through communication and intent links from agent i to agent j , respectively.

Directed link types:

$$e_{j \rightarrow i}^{(t)} \in \{\text{trusted, untrusted, cooperative, deceptive}\}$$

Update rule given event:

$$s_i^{(t+1)} = \begin{cases} f_{\text{obs}}(s_i^{(t)}, E_t) & \text{if observed} \\ f_{\text{comm}}(s_i^{(t)}, E_t) & \text{if trusted} \\ f_{\text{intent}}(s_i^{(t)}, E_t) & \text{if cooperative or deceptive} \\ s_i^{(t)} & \text{otherwise} \end{cases}$$

Related LLM reasoning scaffolds include temporal belief-state chains, as in TimeToM [6], and graph-structured reasoning, as in Graph of Thoughts (GoT) [7]. Temporal Belief Graph combines the temporal dimension with graph representations for multi-agent belief propagation.

Evaluation Metrics

To evaluate temporal belief reasoning, we use four metrics over the predicted graph versus the ground truth.

1. Final Node Accuracy (FNA)

Evaluates accuracy of the final belief state for each agent:

$$\text{FNA} = \frac{1}{|A|} \sum_{i \in A} \mathbf{1}(\hat{b}_i^{(T)} = b_i^{(T)})$$

where A is the set of agents and $b_i^{(T)}$ is agent i 's belief after the final event. FNA evaluates the correct final node state, independent of intermediate reasoning.

2. Time-Respecting Node Accuracy (TNA)

$$\text{TNA} = \frac{\sum_{t=1}^T \sum_{v \in V_t} \mathbf{1}(\hat{b}_v^{(t)} = b_v^{(t)})}{\sum_{t=1}^T |V_t|}$$

where V_t is the node set at timestep t , and T is the number of timesteps. TNA evaluates whether the model's reasoning correctly tracks all intermediate events in the right temporal order, not just the final node state.

3. Temporal edge F1

Each temporal edge is a tuple (source agent, target agent, relation type, time), where relation type $\in \{\text{trusted, untrusted, cooperative, deceptive}\}$. We report F1 over the edge set in two ways:

$$\text{Precision} = \frac{|\hat{E} \cap E|}{|\hat{E}|}, \quad \text{Recall} = \frac{|\hat{E} \cap E|}{|E|}, \quad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Temporal F1 requires full tuple equality. Static F1 requires equality on (source agent, target agent, relation type) only. The difference between temporal and static F1 distinguishes timing errors from topological errors: a model that correctly identifies the edges but misattributes their timestamps will have low temporal F1 and high static F1.

4. Normalized Structural Distance (NSD)

Symmetric distance between predicted and ground truth edge sets, normalized by their union:

$$\text{NSD} = \frac{|\hat{E} \triangle E|}{|\hat{E} \cup E|} = 1 - \text{Jaccard}(\hat{E}, E)$$

where $\hat{E} \triangle E = (\hat{E} \setminus E) \cup (E \setminus \hat{E})$ is the symmetric difference. NSD ranges over $[0,1]$, with lower values indicating higher agreement. Unlike the exact graph match, which equals zero on most non-trivial graphs, NSD provides a continuous comparison. We report the exact TBG reconstruction rate as the fraction of stories with $\text{NSD} = 0$.

Reasoning interpretability. FNA and TNA measure the model’s belief-tracking ability in the final state and intermediate temporal steps. Edge F1 (temporal and static) measures reconstructed graph topology; NSD measures overall graph overlap. A model with high FNA and low edge F1 reaches correct conclusions through unfaithful reasoning; a model with high edge F1 and low FNA constructs correct graph topology but infers the final node states incorrectly. The evaluation contribution is in distinguishing between producing the correct final answer and demonstrating faithful reasoning.

References

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